

Formula SAE

Driverless Systems Form

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DSF - Driverless Systems Form Requirements

DSF.1 Overall

DSF.1.1 Formatting

DSF.1.1.1 Team name, University, and Car Number must be written on every page.

DSF.1.1.2 Document name and subsection name must be written on every page.

Example, "DSF - Actuator Power Supply Schematic"

DSF.1.1.3 All pages must be numbered, including the total number of pages.

Example: "Page 1 of 5"

DSF.1.1.4 Pictures must be well-sized and in good resolution (vector or at least 300dpi).

DSF.1.1.5 Relevant text must be embedded as searchable text and not as a picture.

DSF.1.1.6 All components names, e.g. for sensors or valves, must be used consistently across all submitted documents.

DSF.1.1.7 The document must be printable in US Letter (ANSI A) or DIN A4 formats.

DSF.2 Actuator Power Supply Schematic

The Actuator Power Supply Schematic is used to understand how the Driverless System (DS) and its actuators, Driverless System Brake (DSB) and steering system, are supplied from the LV power source and properly switched via the LVMS and the Driverless System Master Switch (DSMS).

DSF.2.1 Technical Content

DSF.2.1.1 The following must be shown:

- a. LV power source

- b. LVMS and DSMS
- c. Remote Stop System (RSS) bypass relay
- d. Supply of the steering actuator and all electrical brake actuators
- e. A DSMS relay, if used

DSF.2.1.2 The supply of the Driverless System ECUs should be shown on a high level.

DSF.2.1.3 The Battery used for power supply may be abstracted.

DSF.2.1.4 The Emergency Brake System (EBS) supply may be abstracted.

DSF.2.1.5 Provide the following information for the RSS bypass relay in the form section:

- a. Manufacturer
- b. Part Number
- c. Contact certification, proving forcibly guided or mirrored safety contacts.

DSF.2.2 System Requirements Checklist

DSF.2.2.1 Make sure the actuators are properly supplied with power.

DSF.2.2.2 Make sure the RSS is always functional when required, concerning the bypass relay.

DSF.2.2.3 Does the LVMS completely disable the power to the LVS?

DSF.2.2.4 Is the power supply of the steering and braking actuators switched by LVMS and DSMS?

DSF.2.2.5 Is the RSS bypass relay as required by the rules?

DSF.2.2.6 Is the DSMS relay placed as required by the rules?

DSF.3 Continuous Monitoring Flowchart

The Supervision Flowcharts are used to understand the failure detection mechanisms of the Driverless System Brake (DSB).

DSF.3.1 Technical Content

DSF.3.1.1 The document must include a continuous monitoring flowchart.

DSF.3.1.2 The following must be shown in the flowchart:

- a. All system checks performed during operation/driving
- b. The corresponding actions in case of failure
- c. An indication where the supervisor watchdog is handled

DSF.3.1.3 Flow decisions based on sensor data must include the sensor name and the value it is compared to in SI units. The unit “bar” may also be used instead of “Pa”.

DSF.3.1.4 States must be properly named and have clear transitions with annotations if needed.

DSF.3.2 System Requirements Checklist

DSF.3.2.1 Make sure all failure cases are present and handled correctly. Derive test cases that may be used in technical inspection.

DSF.3.2.2 Make sure energy storage and related parameters are checked continuously.

DSF.3.2.3 Does the watchdog operate properly?

- DSF.3.2.4 Is the brake function checked in case of open SDC and is the redundant system triggered in case of malfunction?
- DSF.3.2.5 Is the error reaction as required by the rules (DT.3.2.3)?
- DSF.3.2.6 Are the monitoring thresholds (e.g. brake pressure) as stated in "Pressure System Overview" Form?

DSF.4 High Level Block Diagram

The High Level Block Diagram is used to understand the functionality of your Driverless System Brake (DSB) and to show the interaction of the different DSB parts. It is used to check at a very high level that the system has the structure to fulfill especially the Functional Safety requirements (DT.3.1.3).

DSF.4.1 Technical Content

- DSF.4.1.1 The high level block diagram must include the whole DSB (EBS and its redundancy).
- DSF.4.1.2 All ECUs involved in the DSB function and supervision must be depicted including their watchdogs (if applicable).
- DSF.4.1.3 Non-DSB ECUs involved in the DSB functionality must be shown in the block diagram e.g. if another ECU is used as "watchdog" for the supervisor.
- DSF.4.1.4 The whole energy flow from the energy source to the vehicle brakes must be shown.
- DSF.4.1.5 All connections between electrical, mechanical and hydraulic or pneumatic systems must be shown.
- DSF.4.1.6 All EBS relevant actuators (e.g. valves) must be included and named like in the Pressure System Overview.
- DSF.4.1.7 All DSB relevant sensors must be shown. This includes all signal processing units between the sensor and the supervisor (e.g. analog → CAN converters).
- DSF.4.1.8 The vehicle's Shutdown Circuit must be included but may be abbreviated as SDC.
- DSF.4.1.9 The supply of the EBS actuators must be present (positive as well as GND).
- DSF.4.1.10 The relay used for the EBS activation over the SDC must be named with manufacturer and part number.

DSF.4.2 System Requirements Checklist

- DSF.4.2.1 Make sure the overall system concept is properly built with redundancy.
- DSF.4.2.2 Make sure the RSS is integrated properly.
- DSF.4.2.3 Make sure the EBS supply rules (DT.3.2.1.b) are fulfilled.
- DSF.4.2.4 Make sure the system is sufficient to start a more detailed review and that the diagram provides a good overview to support the other reviewers.
- DSF.4.2.5 Are the document requirements fulfilled and does it fulfill the general style requirements?
- DSF.4.2.6 Is the system function understandable from the block diagram only?
- DSF.4.2.7 Is the EBS supply relay connected correctly (DT.3.2.1.b) and is a proper relay used, not just a semiconductor?
- DSF.4.2.8 Does the system have sufficient sensors to detect failures?

- DSF.4.2.9 Is it possible to test the redundancy independently on startup (different electrical actuation paths)?
- DSF.4.2.10 Does the system show any risks which the team should be aware of, e.g. special material requirements (brake fluid), common release valves which need protection against accidental actuation?

DSF.5 Pressure System Overview

The Pressure System Diagram is used to get an overview of all relevant sensors and actuators of the Driverless System Brake (DSB). All components of the pneumatic and/or the hydraulic system must be present to check the system for single failure reaction capability and to cross-check with the system monitoring of the DSB.

DSF.5.1 Technical Content

- DSF.5.1.1 All parts of the hydraulic and pneumatic system must appear in the document, including sensors. For identical redundant circuits please show both circuits including individual naming.
- DSF.5.1.2 The whole energy flow from the storage down to the brake calipers must be shown.
- DSF.5.1.3 Line connectors don't have to be marked explicitly, if they are used for permanent connection
- DSF.5.1.4 Every component must be named uniquely by function, must have an abbreviation according to EN 81346.
- DSF.5.1.5 Every purchased component must be named with the following format:
< Description >: < Manufacturer > - < Partnumber >, Max. < MaxOperatingpressure >
Example: Compact air cylinder: Festo - ADN-63-40-A-P-A, Max. 10bar
- DSF.5.1.6 Self-built items must be labeled with "self-built" and be clearly identifiable in the mechanical assembly drawings.
- DSF.5.1.7 Pressure regulators must additionally be labeled with its output pressure range.
- DSF.5.1.8 On high pressure equipment a certification shall be stated (e.g. on Paintball equipment).
- DSF.5.1.9 All pressure levels must be annotated in the diagram.
- DSF.5.1.10 If more than one coil per valve is used, each coil must be named uniquely to be clearly identified in the wiring diagram. Either as mentioned in EN 81346 as MB... or by adding an sub index like QM1.1; QM1.2
- DSF.5.1.11 For hydraulic and pneumatic components, symbols according to DIN ISO 1219 should be used.
- DSF.5.1.12 If a pure mechanical system is used to store the energy for emergency braking, the vehicle brake circuit must be shown.

DSF.5.2 System Requirements Checklist

- DSF.5.2.1 Make sure general document requirements are met per DSF.1.1.
- DSF.5.2.2 Make sure that no single points of failure are present in a redundant system.
- DSF.5.2.3 Make sure that master brake cylinders (including pressure transducers) are not connected in series.

- DSF.5.2.4 Make sure sufficient sensors are in the right positions to ensure proper monitoring (e.g. switch of release valves after startup check sequence).
- DSF.5.2.5 Crosscheck pressure ratings with pressure levels and overpressure protection (if applicable). Indicate if the overpressure protection valve is adjustable, if present.
- DSF.5.2.6 Make sure high pressure components have a proper certification.
- DSF.5.2.7 Make sure the volume of pressure tanks is big enough to ensure enough energy is available for EBS maneuver according to DT.3.2 with minimum pressure.

DSF.6 Startup Checks Flowchart

The Supervision Flowcharts are used to understand the failure detection mechanisms of the Driverless System Brake (DSB).

DSF.6.1 Technical Content

- DSF.6.1.1 A flowchart for DSB startup checks must be included.
- DSF.6.1.2 The flowchart must include all actions from DSMS On → AS Ready.
- DSF.6.1.3 Flow decisions, based on sensor data must include the sensor name and the value it is compared to in SI units. The unit “bar” may also be used instead of “Pa”.
- DSF.6.1.4 States must be properly named and have clear transitions with annotations if needed.

DSF.6.2 System Requirements Checklist

- DSF.6.2.1 Make sure all failure cases are present and handled correctly. Derive tests cases that may be used in technical inspection.
- DSF.6.2.2 Are at least the EBS and one redundant system checked to work proper and independently?
- DSF.6.2.3 Is the supervisor watchdog checked to be working?
- DSF.6.2.4 Are the sensors checked for plausibility? E.g. by checking changing values, transfer functions, cross checks or at least by some window comparison?
- DSF.6.2.5 Is the error reaction as required by the rules (DT.3.2.3)?
- DSF.6.2.6 Are the monitoring thresholds (e.g. brake pressure) as stated in “Mechanical Assembly Drawings” Form?

DSF.7 Mechanical Assembly Drawings

The Mechanical Assembly Drawings document is used to understand and make sure your system assemblies. Every part that transfers load inside the Driverless System Brake (DSB) and which could malfunction must be shown here. A typical part is the pedal box assembly or some self-made pressure transducers.

DSF.7.1 Technical Content

- DSF.7.1.1 All DSB relevant assemblies must be shown here.
- DSF.7.1.2 All self-built parts from the Pressure System diagram shown here.
- DSF.7.1.3 Parts and assemblies must be shown as renderings from different perspectives to enable an understanding of the functionality and to determine critical load paths.
- DSF.7.1.4 Multiple views of the same part/assembly shall be on the same page

- DSF.7.1.5 All relevant parts shown must be named uniquely by their function e.g. brake pedal connection rod for more precise feedback requests.
- DSF.7.1.6 Renderings of single parts in an assembly may be provided whenever necessary for a clear understanding.
- DSF.7.1.7 Cut/transparent views must be provided for enclosed assemblies e.g. for self-built hydraulic parts.
- DSF.7.1.8 Only one assembly/standalone part per page. Sub-assemblies or parts of an assembly might be put together onto one or multiple additional pages.
- DSF.7.1.9 If sensors are used to measure mechanical properties (e.g. position, force) they must appear here and be named properly to be clearly identified in all other documents.
- DSF.7.1.10 If a pre-loaded mechanical system is used it must be clear how the system is secured for manual driving.
- DSF.7.1.11 Sealing materials must be annotated for all sealings which may come into contact with brake fluid.

DSF.7.2 System Requirements Checklist

- DSF.7.2.1 Understand the mechanical system construction and identify possible failure points due to insufficient designs.
- DSF.7.2.2 Make sure no lateral forces act on any piston.
- DSF.7.2.3 Make sure all sealing materials are applicable for the fluids they come in contact with.
- DSF.7.2.4 Make sure pre-loaded energy storages can be safely locked in position.

DSF.8 CAN Database File

Teams must submit a .DBC file before the competition to understand and reproduce the system state in case of failure (EBS is activated due to range loss of the RSS). A basic set of DV signals that have to be monitored by the DSB are to be recorded by the energy meter. To be able to evaluate the recorded data, each team needs to provide a DBC file that includes a definition for all the signals mentioned above.

DSF.8.1 Technical Content

- DSF.8.1.1 Refer to the Remote Stop System document on the Event Website for information on what signals are required for submission.

DSF.9.1 Document Format

- DSF.9.1.1 The FSAEOnline website submission form only accepts PDFs for this subsection of the DSF. The recommended method for submission preparation is to paste the DBC contents as plaintext into a word processor and save the document as a PDF.

Visit <https://www.csselectronics.com/pages/can-dbc-file-database-intro> for a helpful reference on the DBC file format.

DSF Examples and Templates

Examples

The Following pages are for reference only and do not include all the information necessary for a complete submission.

Templates

The last two pages of this document are blank templates that may be used for submission. There are two versions provided, one portrait and one landscape. Use of these templates is not required—teams may choose to use another format that satisfies the document format guidelines.

VERSION ""

NS_ :

NS_DESC_
CM_
BA_DEF_
BA_
VAL_
CAT_DEF_
CAT_
FILTER
BA_DEF_DEF_
EV_DATA_
ENVVAR_DATA_
SGTYPE_
SGTYPE_VAL_
BA_DEF_SGTYPE_
BA_SGTYPE_
SIG_TYPE_REF_
VAL_TABLE_
SIG_GROUP_
SIG_VALTYPE_
SIGTYPE_VALTYPE_
BO_TX_BU_
BA_DEF_REL_
BA_REL_
BA_DEF_DEF_REL_
BU_SG_REL_
BU_EV_REL_
BU_BO_REL_
SG_MUL_VAL_

BS_:

BU_: MC IMU

BO_ 67 IMU3_Data: 6 IMU

SG_ w_GyroZ1 : 32|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

BO_ 64 IMU0_Data: 8 IMU

SG_ G_AccelY1 : 16|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

SG_ G_AccelX1 : 0|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

BO_66 IMU2_Data: 4 IMU

SG_w_GyroY1 : 16|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

SG_w_GyroX1 : 0|16@1+ (0.00018,-5.72468) [-5.72468|5.72468] "rad/s" MC

BO_256 SystemStatus: 5 MC

SG_BrakeLightStatus : 32|1@1- (1,0) [0|0] "" Vector__XXX

SG_GoMission : 12|1@0- (1,0) [0|0] "" Vector__XXX

BO_513 BrakeControl: 8 MC

SG_BrakeState : 0|4@1- (1,0) [0|0] "" Vector__XXX

BO_65 IMU1_Data: 6 IMU

SG_G_AccelZ1 : 32|16@1+ (0.0005,-6) [-6|6] "G-Force" MC

BA_DEF_SG_ "GenSigSendType" ENUM

"Cyclic","OnWrite","OnWriteWithRepetition","OnChange","OnChangeWithRepetition","IfActive","IfActiveWithRepetition","NoSigSendType";

BA_DEF_SG_ "GenSigInactiveValue" INT 0 0;

BA_DEF_BO_ "GenMsgCycleTime" INT 0 0;

BA_DEF_BO_ "GenMsgSendType" ENUM

"Cyclic","not_used","not_used","not_used","not_used","Cyclic","not_used","IfActive","NoMsgSendType";

BA_DEF_BU_ "NmStationAddress" HEX 0 0;

BA_DEF_ "DBName" STRING ;

BA_DEF_ "BusType" STRING ;

BA_DEF_DEF_ "GenSigSendType" "Cyclic";

BA_DEF_DEF_ "GenSigInactiveValue" 0;

BA_DEF_DEF_ "GenMsgCycleTime" 0;

BA_DEF_DEF_ "GenMsgSendType" "NoMsgSendType";

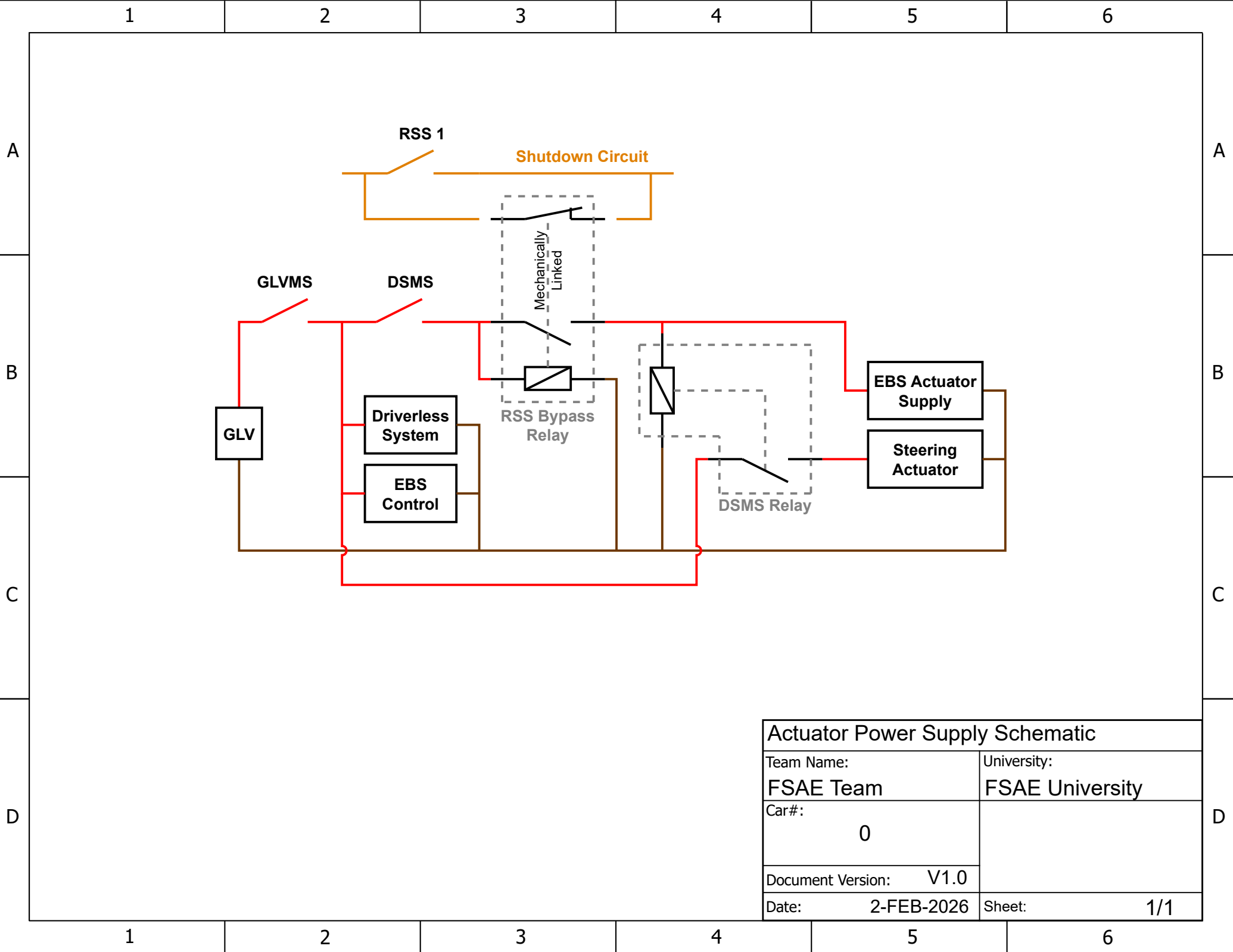
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BA_DEF_DEF_ "DBName" "";

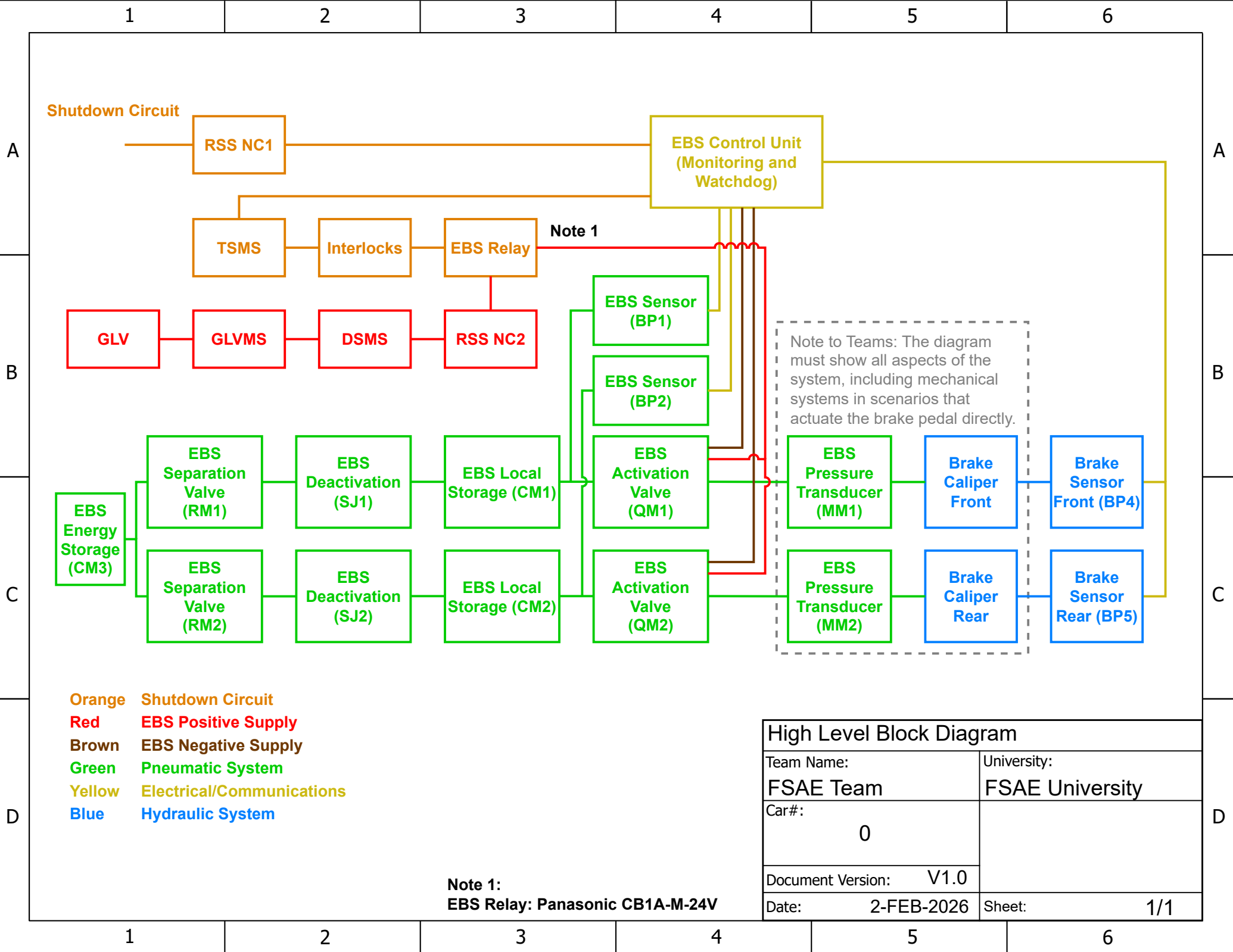
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BA_ "BusType" "CAN";

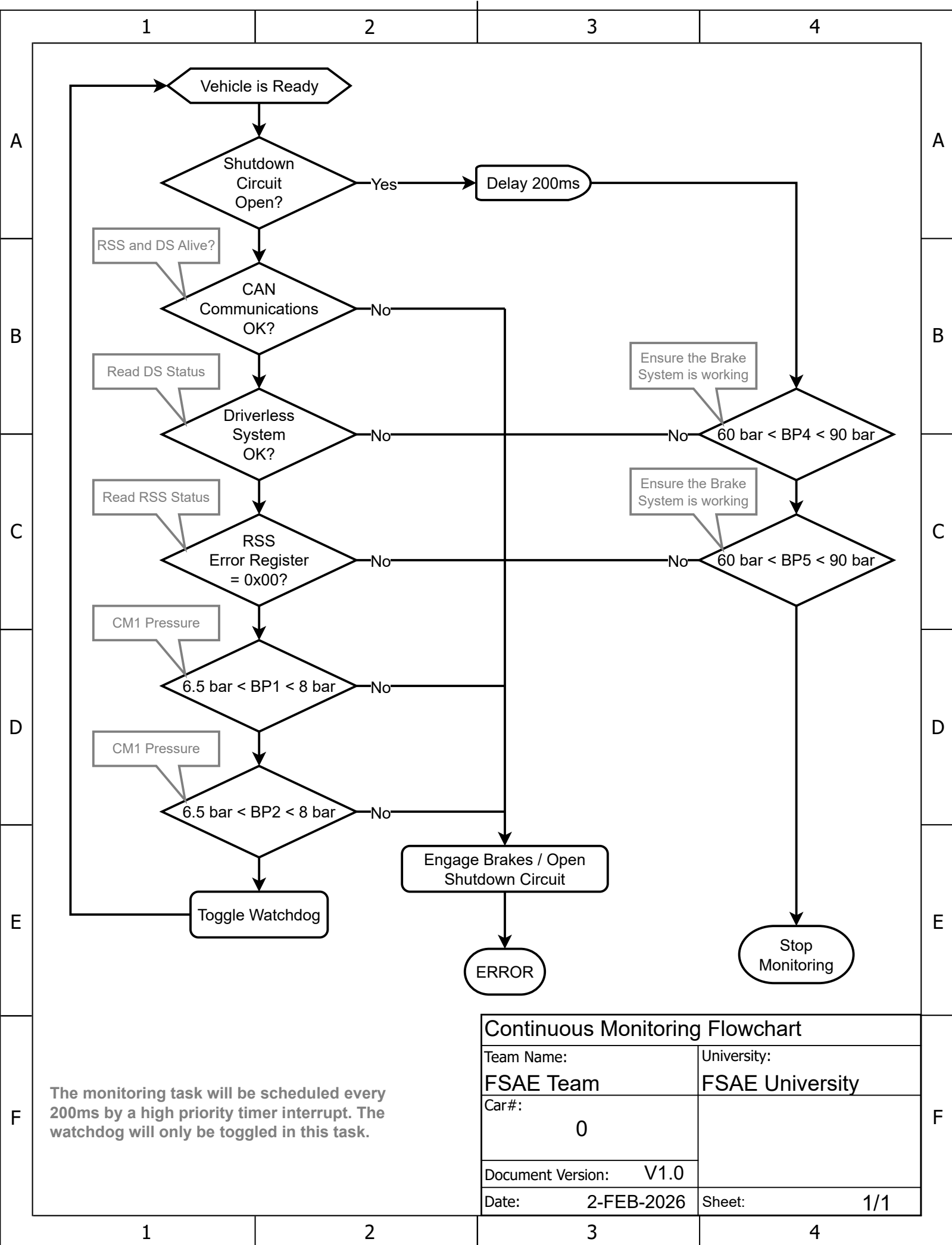
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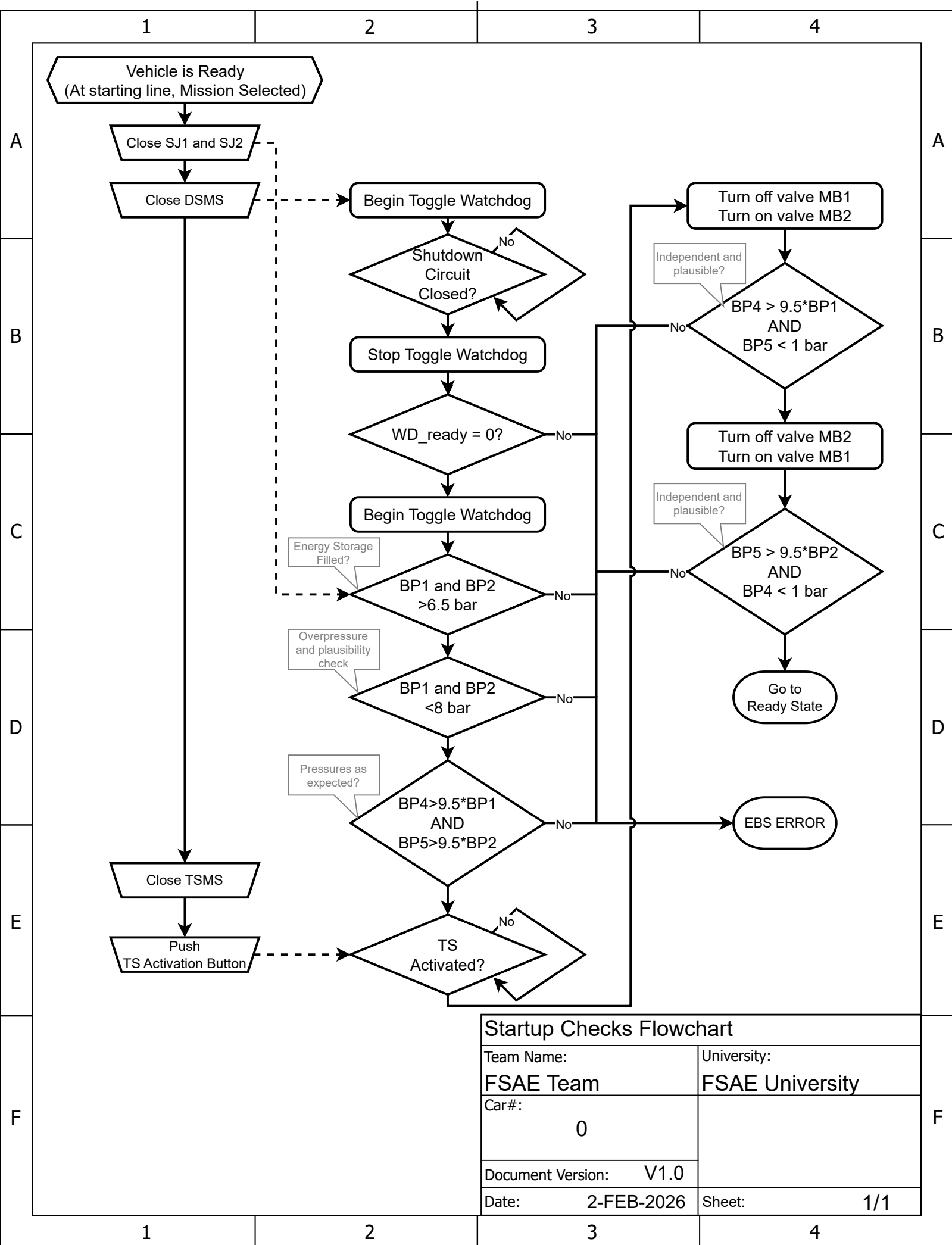


Actuator Power Supply Schematic	
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High Level Block Diagram	
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FSAE Team	FSAE University
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Startup Checks Flowchart

Team Name: FSAE Team

University: FSAE University

Car#: 0

Document Version: V1.0

Date: 2-FEB-2026

Sheet: 1/1

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A

A

B

B

C

C

D

D

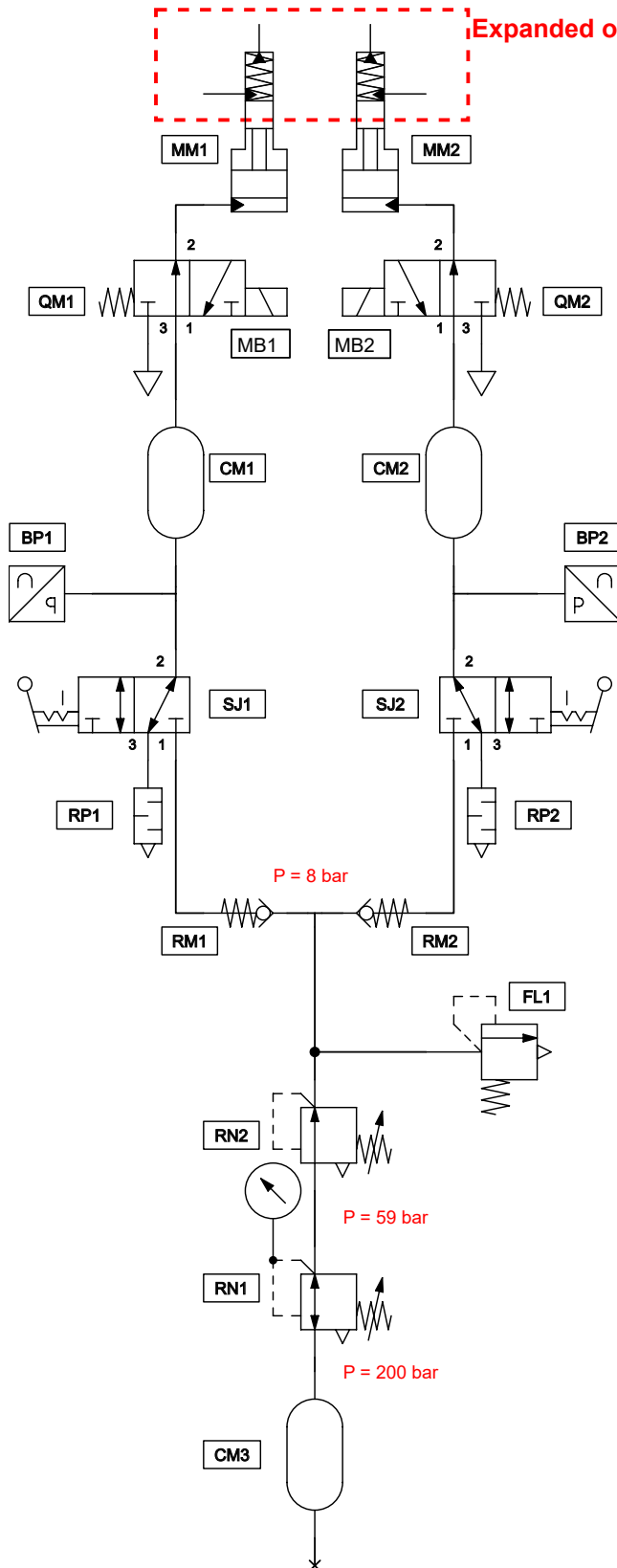
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F

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Expanded on Page 2



EBS Actuator 1 and 2

Pressure Transducer: Self Built - see Mechanical drawings, Max. 10 bar

EBS Activation Valve 1 and 2

Solenoid Valve: FESTO - MHE2-M1H-3/2O-M7, Max. 8 bar

EBS Pressure Tank 1 and 2

Air Tank: FESTO - CRVZS-0.4, Max. 16 bar

EBS Pressure Sensor 1 and 2

Pressure Sensor: FESTO - SPTE-P10R-S6-B-2.5K, Max. 15 bar

EBS Release Valve 1 and 2

Manual Valve: FESTO - H-3-1/4-B 2, Max. 10 bar

Silencer: FESTO - AMTE-M-H-G14, Max. 10 bar

EBS Separation Check-Valve 1 and 2

Check Valve: FESTO - H-1/4-B, Max. 12 bar

Overpressure protection

Relief valve: Landefeld - SV 14 8-8

EBS supply Pressure Regulator

Pressure Regulator: Tescom - BB-13AL3KEA Max. 414 bar

Output Range: 0-15.2 bar

Tank Pressure Regulator

Pressure Regulator: Protoyz - *, Max. 200 bar

Output Range: 59 bar, PI Certified

EBS Supply Pressure Tank

Pressure Tank: Protoyz - 0,8l Alu, Max. 200 bar

PI Certified

Pressure System Overview

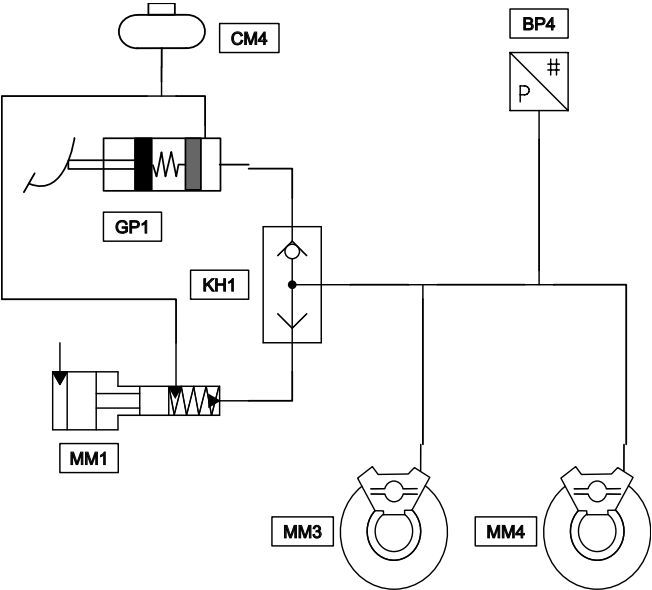
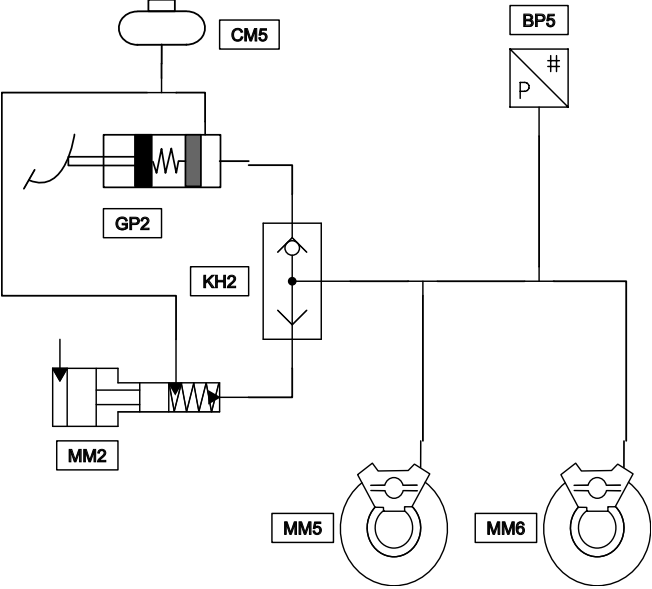
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FSAE Team	FSAE University
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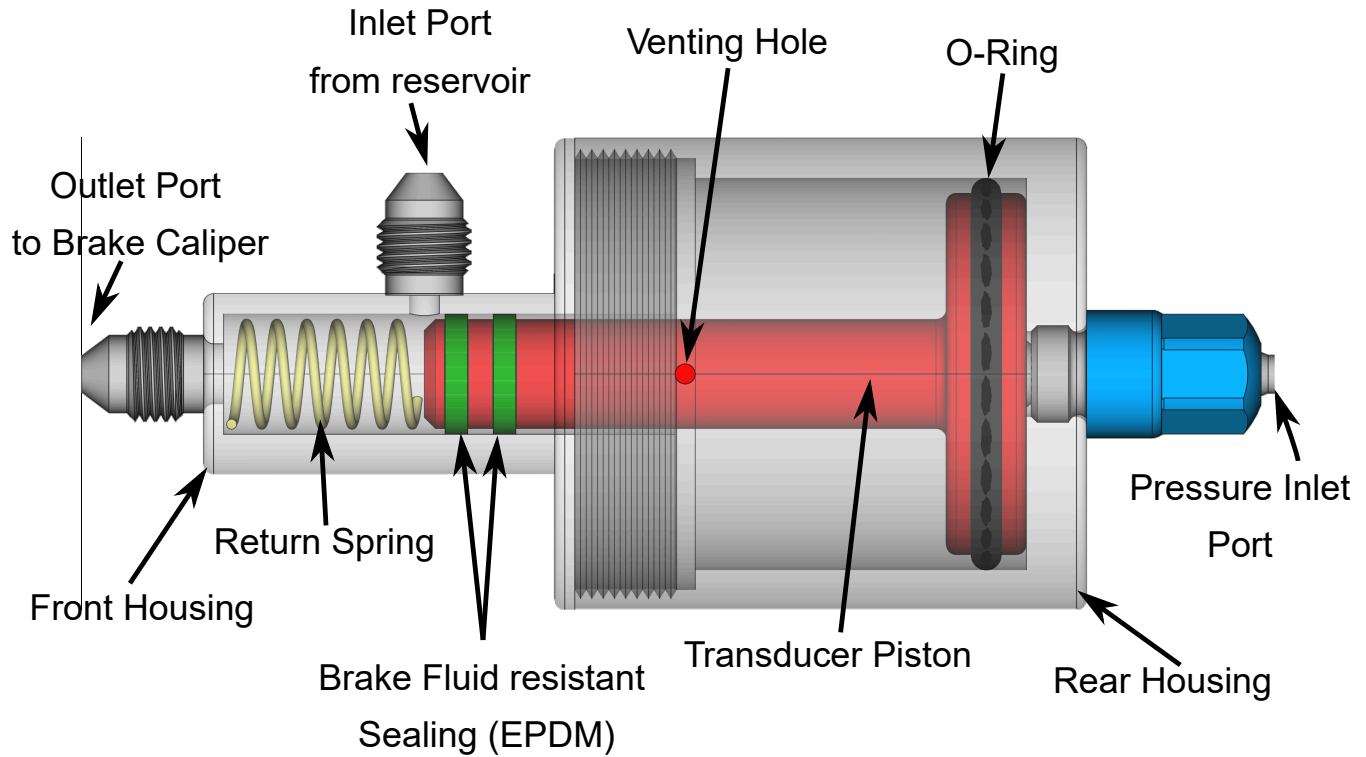
4

	1	2	3	4										
A	Brake Pressure Sensor front Sensor: Variohm - EPT3100 - A - 25000 - B - 5 -B, Max. 250bar Master Cylinder front Master Cylinder: AP racing - CP7854-92PRTE, Max. n.a				A									
B	Shuttle Valve front Shuttle Valve: AP racing - CP5088-1, Max. n.a EBS Actuator 1 Pressure Transducer: Self-built - see Mechanical drawings, Max. 200 bar				B									
C	Front Brake Caliper left & right Brake Caliper: AP racing - CP4227-2S0, Max. n.a				C									
D	Brake Pressure Sensor rear Sensor: Variohm - EPT3100 - A - 25000 - B - 5 -B, Max. 250bar Master Cylinder rear Master Cylinder: AP racing - CP7854-92PRTE, Max. n.a				D									
E	Shuttle Valve rear Shuttle Valve: AP racing - CP5088-1, Max. n.a EBS Actuator 2 Pressure Transducer: Self-built - see Mechanical drawings, Max. 200 bar				E									
F	Rear Brake Caliper left & right Brake Caliper: AP racing - CP4227-2S0, Max. n.a				F									
<table><tr><th colspan="2">Pressure System Overview</th></tr><tr><td>Team Name: FSAE Team</td><td>University: FSAE University</td></tr><tr><td>Car#: 0</td><td rowspan="3"></td></tr><tr><td>Document Version: V1.0</td></tr><tr><td>Date: 2-FEB-2026</td></tr><tr><td>Sheet: 2/2</td></tr></table>					Pressure System Overview		Team Name: FSAE Team	University: FSAE University	Car#: 0		Document Version: V1.0	Date: 2-FEB-2026	Sheet: 2/2	
Pressure System Overview														
Team Name: FSAE Team	University: FSAE University													
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EBS Actuator 1&2 (MM1 / MM2)

d1 = 38 mm, d2 = 12 mm, Transfer Ratio = 1:10

Spring preload = 33 N (0.3 bar pneumatic)



Mechanical Assembly

Team Name:	University:
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B					B
C					C
D					D
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B						B										
C						C										
D						D										
E						E										
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